Statistics & Probability

ANSWER KEY

Permutations

1 How many ways can 5 distinct books be arranged on a shelf?

$5! = 120.$

2 In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 12 members?

${}_{12}P_3 = 12 \times 11 \times 10 = 1320.$

3 How many distinct arrangements are there of the letters in the word APPLE?

There are 5 letters with the letter P repeated twice: $\dfrac{5!}{2!} = \dfrac{120}{2} = 60.$

Combinations

4 How many ways can a committee of 4 be chosen from 11 people?

$\binom{11}{4} = \dfrac{11!}{4!\,7!} = 330.$

5 A pizza shop offers 8 toppings. How many different 3-topping pizzas can be made?

$\binom{8}{3} = \dfrac{8 \times 7 \times 6}{6} = 56.$

6 From a group of 6 men and 5 women, how many ways can a committee of 4 be formed with exactly 2 men and 2 women?

$\binom{6}{2}\binom{5}{2} = 15 \times 10 = 150.$

Probability rules

7 A fair die is rolled. Find $P(\text{even or greater than } 4)$.

Even $= \{2, 4, 6\}$, greater than $4 = \{5, 6\}$. Union $= \{2, 4, 5, 6\}$. $P = \dfrac{4}{6} = \dfrac{2}{3}$.

8 $P(A) = 0.45$ and $P(B) = 0.3$, and A and B are mutually exclusive. Find $P(A \text{ or } B)$.

$P(A \text{ or } B) = 0.45 + 0.3 = 0.75.$

9 A card is drawn from a standard 52-card deck. Find $P(\text{not a face card})$.

There are 12 face cards, so 40 are not face cards. $P = \dfrac{40}{52} = \dfrac{10}{13}$.

Conditional probability and independence

10 $P(A) = 0.5$, $P(B) = 0.4$, $P(A \cap B) = 0.2$. Are A and B independent?

$P(A)P(B) = 0.5 \times 0.4 = 0.2 = P(A \cap B)$. Yes, A and B are independent.

11 In a class, 60% of students play sports, and 25% of all students both play sports and are honor students. Find $P(\text{honor student} \mid \text{plays sports})$.

$P(\text{honor} \mid \text{sports}) = \frac{0.25}{0.6} \approx 0.417$, or about 41.7%.

12 Events C and D are independent, with $P(C) = 0.7$ and $P(D) = 0.6$. Find $P(C \cup D)$.

$P(C \cap D) = 0.7 \times 0.6 = 0.42$. $P(C \cup D) = 0.7 + 0.6 - 0.42 = 0.88$.

Measures of center and spread

13 Find the mean and median of the data set $12, 15, 18, 20, 25$.

Mean $= \frac{12 + 15 + 18 + 20 + 25}{5} = \frac{90}{5} = 18$. Median (middle value, already sorted) $= 18$.

14 Find the population standard deviation of the data set $4, 8, 6, 5, 3$.

Mean $= \frac{26}{5} = 5.2$. Squared deviations: $1.44, 7.84, 0.64, 0.04, 4.84$, summing to 14.8. Variance $= \frac{14.8}{5} = 2.96$. Standard deviation $= \sqrt{2.96} \approx 1.72$.

Data representation

15 The bar chart shows the distribution of grades in a class of 37 students. Find the probability that a randomly selected student earned a grade of A or B, as a percent to the nearest whole percent.

$P(A \text{ or } B) = \frac{8 + 15}{37} = \frac{23}{37} \approx 0.622$, or about 62%.

16 Using the same bar chart, find the probability that a randomly selected student scored a C or D grade, as a percent to the nearest whole percent.

$P(C \text{ or } D) = \frac{10 + 4}{37} = \frac{14}{37} \approx 0.378$, or about 38%.

The binomial theorem

17 Find the coefficient of x^3y^2 in the expansion of $(x + y)^5$.

The term is $\binom{5}{2}x^3y^2$, so the coefficient is $\binom{5}{2} = 10$.

18 In the expansion of $(2x - 1)^4$, find the coefficient of x^2.

The general term is $\binom{4}{k}(2x)^{4-k}(-1)^k$. For x^2, $4 - k = 2 \Rightarrow k = 2$: $\binom{4}{2}(2x)^2(-1)^2 = 6(4x^2) = 24x^2$. The coefficient is 24.

The normal distribution

19 A data set is normally distributed with mean 100 and standard deviation 15. What percent of values lie between 70 and 130?

70 and 130 are each 2 standard deviations from the mean (100 ± 30). By the 68-95-99.7 rule, about 95% of values lie in this range.

20 For the same distribution (mean 100, standard deviation 15), find the approximate percentage of values greater than 115.

115 is 1 standard deviation above the mean. About 68% of values lie within 1 standard deviation, so 34% lie between the mean and 115. Since 50% lie above the mean, the percentage above 115 is $50\% - 34\% = 16\%$.

A well-designed word problem: quality control sampling

21 This question has three parts. A factory produces batteries, and 4% of batteries are defective. A quality-control inspector randomly selects a sample of 3 batteries from a large shipment (assume the selections are independent). (a) Find the probability that none of the 3 batteries are defective. (b) Find the probability that exactly 1 of the 3 batteries is defective. (c) Find the probability that at least 1 of the 3 batteries is defective.

(a) $P(\text{none defective}) = (0.96)^3 = 0.884736 \approx 88.5\%$. (b) $P(\text{exactly } 1) = \binom{3}{1}(0.04)(0.96)^2 = 3(0.04)(0.9216) = 0.110592 \approx 11.1\%$. (c) $P(\text{at least } 1) = 1 - P(\text{none}) = 1 - 0.884736 = 0.115264 \approx 11.5\%$.